

MG-HGNN: Modality-Gated Heterogeneous Graph Neural Networks for Student Academic Performance Prediction

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ABSTRACT

Predicting student academic performance is a critical task in educational data mining, enabling early identification of at-risk learners and timely pedagogical intervention. Recent advances in graph neural networks (GNNs) have demonstrated strong potential by explicitly modelling relational structures among students, courses, and learning activities. However, existing heterogeneous GNN (HGNN) approaches share a fundamental limitation: they collapse all available modalities — structured academic records, textual content, and behavioral interaction logs — into a single, unified node embedding. This flattening ignores the fact that different edge relation types inherently carry different modal signals; for instance, a student’s behavioral clickstream is more informative when propagating messages through a *resource-access* edge than through a *peer-collaboration* edge, where textual communication patterns are dominant.

In this paper, we propose **MG-HGNN**, a *Modality-Gated Heterogeneous Graph Neural Network* that addresses this gap. Each node in the graph maintains three *separate* modality embeddings — a structured embedding produced by an MLP encoder, a textual embedding from a fine-tuned BERT encoder, and a behavioral embedding from a bidirectional LSTM encoder. The core innovation is a lightweight, learnable *modality gate*: for each edge relation type r , a dedicated softmax gate $\alpha^{(r)} = [\alpha_s^{(r)}, \alpha_t^{(r)}, \alpha_b^{(r)}]$ is learned end-to-end, dynamically controlling how much each modality contributes to message passing along that specific relation. The gate mechanism is conditioned exclusively on the relation type, imposing no additional overhead at the node level.

We evaluate MG-HGNN on two publicly available educational datasets — MOOC-Cube and the Open University Learning Analytics Dataset (OULAD) — across three complementary prediction tasks: final grade regression, dropout risk classification, and engagement-level prediction. Experimental results show that MG-HGNN consistently outperforms strong baselines including HAN, HGT, R-GCN, and multi-topology GNN variants. Ablation studies confirm that edge-conditioned gating is the key factor driving the performance gain, and post-training analysis of learned gate

weights validates our core hypothesis: the model autonomously discovers that behavioral signals are most relevant for resource-access edges, textual signals for collaboration edges, and structured signals for enrollment edges — without any manual specification of these inductive biases.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; *Artificial intelligence*.

KEYWORDS

heterogeneous graph neural networks, student performance prediction, multi-modal learning, modality gating, educational data mining, learning analytics

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1 INTRODUCTION

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2 RELATED WORK

The prediction of student academic performance has evolved substantially over the past decade, progressing from classical machine learning models operating on independent per-student feature vectors toward relational deep learning methods that explicitly encode the complex network structure of educational ecosystems. This section surveys the most relevant families of approaches, situating our proposed MG-HGNN within the existing landscape.

2.1 Graph Neural Networks for Student Performance Prediction

Early graph-based approaches demonstrated that modelling student similarity as an explicit graph structure yields systematic gains over flat-feature baselines. (author?) [4] proposed Study-GNN, a multi-topology GNN (MTGNN) in which multiple student similarity graphs are constructed using distinct distance metrics. An attention mechanism then fuses these topologies, casting performance prediction as semi-supervised node classification where each node represents a student and labels encode pass, fail, or withdrawal outcomes. Study-GNN was shown to outperform support vector machines, logistic regression, shallow neural networks, and vanilla graph convolutional networks, validating the utility of multi-view relational modelling.

Building on the insight that student-student similarity alone is insufficient, (author?) [2] introduced APP-DGNN, a dual graph neural network that constructs two complementary graphs: one derived from interaction logs capturing engagement dynamics over time, and another from student attributes such as demographics and academic background. Local representations from the interaction graph and global representations from the attribute graph are fused for downstream prediction, achieving approximately 84% accuracy on pass/fail classification and approximately 90% on pass/withdraw tasks on a public MOOC dataset. This dual-graph paradigm highlights a key architectural principle: different information sources warrant distinct graph topologies rather than a single aggregated representation.

2.2 Temporal and Dynamic Graph Models

A recurring limitation of static graph methods is their inability to capture the evolution of student behavior and performance across time. (author?) [3] addressed this with APP-TGN, a temporal graph network (TGN) in which student activity logs are encoded as a dynamic graph where edges carry timestamps. The model combines low-pass and high-pass spectral filters with global neighborhood sampling and multi-head attention, significantly improving both overall predictive accuracy and early at-risk detection in massive open online course (MOOC) settings. Similarly, (author?) [7] demonstrated the value of incorporating temporal context within a richer hybrid architecture, embedding GNN layers inside a GNN-Transformer-InceptionNet pipeline for multi-label student performance prediction, reporting up to 98.5% accuracy on a single-institution dataset.

Despite these advances, several dual and multi-graph designs still do not fully exploit long-range temporal dependencies [1–3], motivating continued work on temporally expressive architectures. Our proposed MG-HGNN addresses a complementary and previously

unexplored axis: rather than improving temporal modelling, we improve *modal selectivity* during heterogeneous message passing.

2.3 Hypergraph and Higher-Order Models

Beyond pairwise edges, (author?) [6] proposed FD-HGNN, a framelet-based dual hypergraph neural network for student performance prediction. Framelet transforms are used to produce low-pass and high-pass hypergraphs over students, and a dual-channel attention mechanism models higher-order, multi-way relationships among student groups rather than individual pairs. Evaluated across four datasets, FD-HGNN outperforms both traditional machine learning methods and standard GNNs. This line of work underscores the importance of capturing structural complexity beyond simple pairwise student-course relations; however, hypergraph constructions do not directly address the problem of multi-modal feature heterogeneity within each node.

2.4 Multi-source and Heterogeneous Data Fusion

Several works have emphasised the need to handle sparse, multi-source data including behavioral logs, demographic features, emotional indicators, and prior academic records more effectively than flat concatenation allows [1–4, 6, 7]. (author?) [1] proposed the Multi-Dimensional Student Performance Prediction model (MSPP), which incorporates advanced graph neural network layers on heterogeneous academic data, achieving approximately 76% accuracy on complex multi-class classification tasks. This work demonstrates that even with heterogeneous input, current methods still rely on pre-fusion of modalities before graph operations, limiting the expressiveness of relational message passing.

Collectively, these contributions establish graph-based deep learning as the dominant paradigm for student performance modelling, with reported accuracy ranging from approximately 76% on complex multi-class tasks [1] to 84–90% on binary outcomes [2] and up to 98–99% in richer, typically single-institution settings [7]. Shared limitations across this body of work include high computational cost for complex GNN operators on large student cohorts, and questions about cross-institutional generalizability [1–4, 6, 7].

2.5 Gap Addressed by This Work

Despite the breadth of existing approaches, a fundamental assumption has remained unexamined: that all relation types in the educational graph should draw on the same mixture of input modalities when aggregating neighborhood information. No existing HGNN framework conditions its feature fusion strategy on the type of edge being traversed. This is the precise gap that MG-HGNN fills. By maintaining separate per-modality embeddings at each node and learning a distinct, softmax-normalized modality gate $\alpha^{(r)}$ for each edge relation type $r \in \mathcal{R}$, MG-HGNN allows the graph itself to discover which information sources are most relevant for each relational context — without requiring manual specification of these inductive biases.

3 PROBLEM DEFINITION

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4 METHODOLOGY: MG-HGNN

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5 EXPERIMENTS

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6 CONCLUSION

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